

## **Development Choices**

For this project, I focused on building a 3D scene using low-polygon objects constructed from basic primitives such as cubes, cylinders, spheres, and planes. The chosen method enabled me to create an accurate 2D image structure because I maintained the polygon count below the established threshold. I used multiple primitive shapes to create complex objects instead of creating detailed meshes. The object in my scene used a cylinder and a sphere as two separate components which showed that composite modeling can create better visual results through a method that needs less complex design work.

I used textures in specific areas to create realistic effects. The use of high-quality royalty-free textures enabled me to create detailed surface appearances which maintained their original form. The final presentation depended on the use of appropriate lighting. I used point and directional lights together with Phong shading system components which include ambient and diffuse and specular lighting systems. The system created depth perception and realistic effects which allowed viewers to see all objects from various camera perspectives.

## **Navigation**

Users can freely navigate my 3D scene through all three axes because my navigation system enables complete movement. The keyboard controls use W and S keys for forward and backward camera movement while A and D keys control left and right movement and Q and E keys handle vertical movement. The system enables users to navigate complete spatial dimensions of the virtual environment.

The mouse provides users with two functions because they can use it to move through space and they can use it to rotate the camera view. The mouse movement enables users to control camera angles through pitch and yaw which creates a natural looking-around experience

for them because they remain in one place. I developed scroll wheel functionality which enables users to change their movement speed through scene navigation and narrow spaces.

The system provides users with the option to switch between perspective and orthographic projections through a designated keyboard command. Users can transition between a realistic three-dimensional display and a two-dimensional flattened view which maintains the same camera position for both views, which shows users multiple ways to see the scene.

### **Custom Functions**

The custom functions I created for my code base divided its main tasks into separate sections which enhanced both its organization and its maintainability. The system handles input through an exclusive function which processes both keyboard and mouse activities. The main rendering loop maintains its focus on drawing operations because of this system design which prevents any other tasks from entering the loop.

I developed functions that handle both camera movement and camera orientation changes. The functions create reusable components which let users modify the logic that controls pitch and yaw and their movement speed. The camera system I developed allows me to use it in other projects because its entire system functions remain intact.

I applied modular design techniques to both object rendering and transformation application processes. The transformation functions which include translation and rotation and scaling exist as reusable components which help me position objects accurately throughout the scene. The modular system design enhances document clarity while also simplifying error detection work and future system improvements.